Total No. of Questions : 10]

SEAT No. :

**P3114**

[Total No. of Pages : 2

# [5461]-150-B

## B.E. (Mechanical Engineering)

## DESIGN OF PUMPS, BLOWERS AND COMPRESSORS

## (2012 Course) (End Sem.) (402050 C) (Elective-IV) (Semester-II)

*Time : 2½ Hours]* *[Max. Marks :70*

***Instructions to the candidates:***

1) *Solve Q. 1 or Q. 2, Q. 3 or Q.4, Q. 5 or Q. 6, Q. 7 or Q. 8, Q. 9 or Q. 10.*
2) *Use only one answer book & supplement if required.*
3) *Use of scientific calculator, steam table, mollier chart is allowed.*
4) *Figures to right indicate full marks.*



***Q1)*** a) Write note on: **[5]**

i) Stage Velocity Triangles

ii) Specific Speed

b) Explain performance characteristics curves for fan and blowers. **[5]**

OR

***Q2)*** a) Differentiate between compressible & incompressible flow machines. **[5]**

b) Explain the basic equation of energy transfer between fluid & rotor. **[5]**

***Q3)*** a) Explain with neat sketch of Air vessel in reciprocating pump. **[5]**

b) A double acting reciprocating pump running at 40 rpm is discharging 1 $m^3$ of water per min The pump has a stroke of 400 mm. The diameter of the piston is 200 mm. The delivery & suction head are 20 m & 5 m respectively. Find the slip of the pump & power required to drive the pump. **[5]**

OR

***Q4)*** a) What is slip in pump? Explain the meaning of negative slip. **[5]**

b) A single acting reciprocating pump has a plunger diameter 250 mm & stroke of 450 mm runs at 60 rpm. The length & diameter of delivery pipe are 60 m & 100 mm respectively. Determine the power saved in overcoming friction in the delivery pipe by fitting an air vessel on the delivery side of the sump. Assume friction factor = 0.01. **[5]**

***Q5)*** a) Explain performance of axial fan with graph. **[8]**

b) How the dust erosion of centrifugal impeller does occur? What is its effect on the performance? **[8]**

OR

*Q6)* a) Enlist the methods to reduce the fan noise. **[8]**

b) Discuss various applications of fans & blowers. **[8]**

*Q7)* a) Explain UGV & DGV with velocity triangle. **[8]**

b) A centrifugal blower takes in 180 $m^3$/min of air at suction pressure of 1.013 bar & temperature of 430 C and delivers it at 750 mm of W.G. taking the efficiencies of the blower and drive as 80% & 82% respectively. Determine the power required to drive the blower and the state of air at exit. **[8]**

OR

*Q8)* a) Write short note on selection of blowers for a desired application. **[8]**

b) A centrifugal fan has the following data: **[8]**

| | | |
|---|---|---|
| Inner diameter of the impeller | = | 18 cm |
| Outer diameter of the impeller | = | 20 cm |
| Speed | = | 1450 rpm |
| Relative velocity at entry | = | 20 m/s |
| Absolute velocity at entry | = | 21 m/s |
| Relative velocity at exit | = | 17 m/s |
| absolute velocity at exit | = | 25 m/s |
| Flow rate | = | 0.5 kg/s |
| Motor efficiency | = | 78% |
| Density of air | = | 1.25 kg/$m^3$ |

Determine:

i) Stage pressure rise.

ii) Degree of reaction.

iii) Power to drive the fan.

*Q9)* a) Explain enthalpy-entropy diagram for centrifugal compressor. **[8]**

b) Draw velocity triangles at the entry & exit for the following axial compressor stage. **[10]**

i) R = ½ ii) R < ½

OR

*Q10)* a) What are the basic design features in axial flow compressor? **[8]**

b) Draw & explain performance curves of centrifugal compressors. **[10]**

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